Yumi Heo

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Data Scientist and AI/ML Engineer

I am a data scientist and AI/ML engineer who builds models and takes them into production. At AITIS I build computer vision models for skin cancer and wound care and fine-tune LLMs on clinical data. At the Office of Health Economics I built an LLM pipeline that cut data collection from 2 days to 10 minutes.

SKILLS

ML & Computer Vision: PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, ONNX, Pandas, NumPy
LLM & GenAI: OpenAI API, Claude API, HuggingFace, LLM fine-tuning, prompt engineering, LangChain, LangGraph, Chroma, MCP
Engineering & MLOps: Python, SQL, R, Git, Docker, FastAPI, unit testing, MLflow, PostgreSQL, Jira
Data & Cloud: AWS, GCP, BigQuery, PySpark, Tableau, Streamlit

PROJECTS

- TechNews-briefing: Built a daily email briefing of AI and tech news from the UK, US and South Korea. An OpenAI model in a LangGraph workflow picks the stories, writes the summaries and reviews them, and it runs every morning on GitHub Actions
- NoteGuard: Built a tool that finds and replaces personal information in clinical notes, so they can be used safely for modelling
- Automated-EMA-NICE-Data-Scraping: Built a GPT-4o mini pipeline that summarises regulatory data from EMA and NICE
- Inventory-Placement-Optimiser: Built a supply chain tool that forecasts demand and decides where to keep stock to cut costs

WORK EXPERIENCE

Data Scientist, AITIS, London, UK **Jan 2025 - Present**

AITIS develops clinical AI technology and health-monitoring wearables. I work closely with doctors and dermatologists.

- Developed skin cancer classification models in PyTorch (EfficientNet) on 10,000+ dermatology images, reaching over 90% sensitivity and specificity on a held-out test set and an independent client validation set. The models power the AITIS Skin Cancer app, one of six winners of the Healthcare InnoMatch
- Fine-tuned Phi-3-mini with LoRA on image embeddings via a LLaVA-style projection layer to predict chronic wound healing outcomes from image sequences, using real-world hospital data from Singapore
- Cut clinician annotation workload by 90% with a tool that sends only the images the model is least sure about for expert review
- Worked on signal processing and ML models for wearables that estimate blood pressure, heart rate and glucose
- Deployed models as FastAPI services in Docker with unit tests and tracked experiments in MLflow, working in a cross-functional team with software engineers
- Plan new image datasets with doctors and dermatologists and write the annotation guidelines for them
- Explain model results to non-technical clinical clients through reports and meetings, and presented the skin cancer project at the Data Bites event, City St George's, University of London
- Mentored summer interns during their onboarding

Data Science Intern, Office of Health Economics, London, UK **Jul 2024 - Oct 2024**

Independent health economics research organisation. I built an automated data collection tool for health economists.

- Built an automated LLM pipeline (GPT-4o mini) that collects regulatory data from NICE and the European Medicines Agency, cutting health economists' data collection time from 2 days to 10 minutes
- Engineered a RAG pipeline with Chroma and LangChain to improve retrieval and summarisation accuracy on medical documents
- Applied prompt engineering and ROUGE-based evaluation to iterate on prompts and raise summarisation quality
- Ran hands-on training so non-technical colleagues could run the workflow on their own

Acquisition & Distribution Assistant Manager, Danal Entertainment, South Korea **Apr 2019 - Oct 2022**

Film and TV distributor in South Korea. I managed the distribution of 100+ films and TV series across Netflix, Google Play and Naver.

- Developed statistical forecasting models to predict film profitability across a 10,000+ film catalogue and achieved 45-65% recoupment of licence fees at or before release
- Used forecast insights to negotiate licensing agreements with 50+ international providers, securing an exclusive distribution partnership with Coupang Play, and reported sales performance to senior management

EDUCATION

MSc Data Science (Distinction), City St George's, University of London, UK **Sep 2023 - Jan 2025**

- Dissertation (72%): Developed an interpretable ensemble anomaly detection framework for industrial control systems using DBSCAN labelling, LSTM-GRU stacking and SHAP/LIME explainability
- Modules: NLP (88.5%), Machine Learning (81%), Big Data (78%), Neural Computing (76%), Visual Analytics (75%)

BA Economics and English Language & Literature (2:1), Dongguk University, Seoul, South Korea **Mar 2015 - Feb 2018**